

$$\textcircled{1} f(x) = x^2 - 6x + 5$$

$$x_0 = 0$$

$$\alpha = 0.1$$

$$f'(x) = 2x - 6$$

$$\begin{aligned} x_1 &= x_0 - (0.1)(2 \cdot 0 - 6) \\ &= 0 + (0.6) = \underline{\underline{0.6}} \end{aligned}$$

$$\boxed{x_1 = 0.6}$$

$$x_2 = 0.6 - (0.1)(2 \cdot (0.6) - 6)$$

$$\begin{aligned} x_2 &= 0.6 - (0.1)(1.2 - 6) \\ &= 0.6 + 0.48 \end{aligned}$$

$$\boxed{x_2 = \underline{\underline{1.08}}}$$

$$\textcircled{2} \quad f(x, y) = 2x^2 + y^2$$

$$\frac{\partial f}{\partial x} = 4x$$

$$\frac{\partial f}{\partial y} = 2y$$

$$x_0 = 1 \quad \alpha = 0.2$$

$$y_0 = -2$$

$$x_1 = x_0 - (0.2)(4 \cdot 1) = 1 - 0.8 = \underline{\underline{0.2}}$$

$$y_1 = -2 + (0.2)(2 \cdot -2) = -2 + 0.8 = -1.2$$

$$\underline{\underline{(0.2, -1.2)}}$$

$$\textcircled{8} \quad f(x) = ax^2 \quad a > 0$$

$$f'(x) = 2ax$$

$$x_n = x_{n-1} - \alpha(2ax)$$

$$x_n = x_{n-1} - (2a\alpha)x_{n-1}$$

$$x_n = x_{n-1}(1 - 2a\alpha)$$

A geometric progression $\Rightarrow$

$$x_n = x_0(1 - 2a\alpha)^n$$

④ x_n long run behaviour

$$x_n = x_0 (1 - 2a\alpha)^n$$

a) $0 < 2a\alpha < 2$

$0 > -2a\alpha > -2$

$$\boxed{1 > 1 - 2a\alpha > -1} \rightarrow \text{convergent to zero as } n \rightarrow \infty$$

b) $\alpha = \frac{1}{a}$

$$1 - 2 \cdot \frac{1}{a} \cdot \frac{1}{a} = (-1)^n$$

$x_n \rightarrow$ oscillate between

$$\boxed{x_0 \text{ \& - } x_0}$$

$$c) \quad \alpha > \frac{1}{a}$$

$x_n \rightarrow \infty$ diverges

⑧ It will land on minima when $(1 - 2a\alpha) = 0$

$$\boxed{\alpha = \frac{1}{2a}}$$

This would be a special case of a) where the series equals to zero & instead of converging instantly.

⑥

∇f

$\hookrightarrow -ve$

$\rightarrow$

decreasing

$\rightarrow$

$-\nabla f \rightarrow +ve$

$\downarrow$
moves to
lesser value

$\hookrightarrow +ve$

$\rightarrow$

increasing

$\rightarrow$

$-\nabla f \rightarrow -ve$

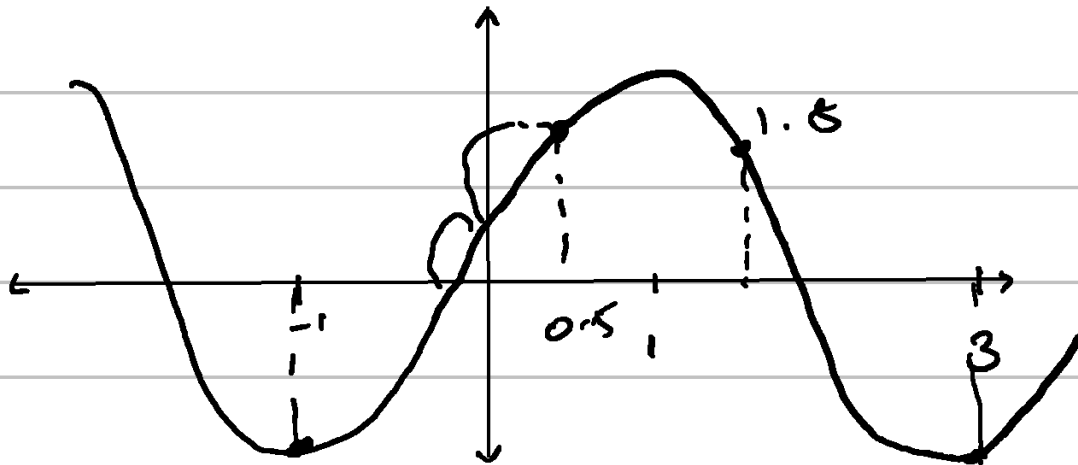
$\uparrow$
moves to
higher value

So, $-\nabla f$ moves opposite to the direction of increasing value i.e. decreasing value.

$$\textcircled{7} E(x) = x^4 - 4x^3 - 2x^2 + 12x + 3$$

Global minima $\rightarrow -1, 3$

Local maxima $\rightarrow 1$



$\Rightarrow$ Since the points 0.5 & 1.5 lie on the either side of maxima they will move in the negative direction of slope, taking $0.5 \rightarrow 1$
 $1.5 \rightarrow 1$